

Ng Kok Jing

harryngkokjing@gmail.com • +65 85040648 • [linkedin.com/in/ngkokjing](https://www.linkedin.com/in/ngkokjing) • harryngkokjing.com

EDUCATION

Singapore Management University

August 2024 - Present

Bachelor of Science (Software Engineering)

Singapore

- Teaching Assistant - Python / Data Structures and Algorithms / Object-Oriented Programming (Java)
- Teaching Assistant - Web Application Development (Express.js & MongoDB)

EXPERIENCE

Software Engineer Intern | Java, Spring Boot, Docker, Kubernetes

May 2026 - July 2026

Eastern International University

Binh Duong, Vietnam

- Migrated infrastructure from Docker Compose to **k3s** cluster on **Ubuntu VPS**, enabling horizontal scaling of production services.
- Implemented **Terraform** configurations to provision **AWS ECR** repositories with custom lifecycle policies.
- Integrated **GitOps** pipeline with **ArgoCD Core**, enabling automated Kubernetes deployments from Git commits.
- Restructured **CI/CD** pipelines to build Docker images, push to AWS container registries, run dependency scanning, and gate deployments on unit test success.
- Deployed a GitOps-managed **Prometheus** and **Grafana** monitoring stack on a k3s cluster using **Helm** and **ArgoCD**.
- Configured **Prometheus Alertmanager** to deliver email and Telegram channel alerts for critical infrastructure events.
- Refactoring fragmented email services into 1 single service with **Thymeleaf** templates, decoupling HTML from business logic.
- Migrating report generation from synchronous to async processing using **Redis Streams**, improving API response times.

Software Engineer Intern | Python, PostgreSQL, RAG, GCP, LangChain

December 2025 - March 2026

AI Shophouse

Singapore

- Built a **LangChain-powered** AI tutor chatbot implementing a full RAG pipeline by ingesting PDFs, chunking documents, generating embeddings via **text-embedding-ada-002**, and storing vectors in **pgvector** on **GCP PostgreSQL**, with top-k similarity search, re-ranking, and prompt templating to deliver contextually accurate responses.
- Integrated **G-EVAL**-based validation to systematically assess and refine chatbot outputs, increasing response quality.

TECHNICAL PROJECTS

FoodIT | Coursework | React Native, Expo Go, FastAPI, Golang, WebSockets, AWS, Apache Kafka, Terraform, Kubernetes | [GitHub](#)

- Architected food delivery platform using **9** microservices (**Go/Python**), enabling independent deployment and horizontal scaling
- Implemented **OAuth** with **AWS Cognito** and **Lambda triggers**, auto-provisioning user records in **DynamoDB** on signup.
- Built **WebSocket** chat service in **Go** with **Cassandra** and **Redis pub/sub (AWS ElastiCache)**, achieving sub-100ms peer messaging.
- Developed real-time GPS tracking via **WebSocket** and **Redis**, broadcasting location updates every 3 seconds.
- Orchestrated async order workflows with **Kafka (AWS MSK)**, decoupling payments, chat, and location services via events.
- Wrote a **GitLab CI/CD** pipeline with SAST scanning, secret detection, Docker image builds, and automated pushes to **AWS ECR**.
- Deployed microservices to **AWS EKS Fargate** via **Terraform and Docker**, eliminating server management overhead.

Sea Link Finance | NUS FinTech Summit 2026 | Next.js, TypeScript, Tailwind CSS, XRPL.js, CrossMark SDK | [DevPost](#), [GitHub](#)

- Built a XRPL maritime finance MVP leveraging **Native Hooks** and **RLUSD** for tokenized voyage financing, placing **3rd place** in the main track at NUS FinTech Summit 2026 by solving shipowner liquidity challenges.
- Developed a transaction monitoring system using XRPL memo polling, incremental sync, and UI reconciliation to deliver live settlement updates without manual refresh.
- Built wallet authentication using **Crossmark SDK** to retrieve addresses and sign XRPL transactions, enabling seamless DID creation and credential onboarding with user approval flows.

Git Gram | Personal Project | Golang, Fiber, GitLab CI/CD, Docker, Google Cloud | [Telegram Link](#), [GitLab](#)

- Built Telegram bot handling GitHub **webhooks** with HMAC verification and **JWT-authenticated** API calls, processing custom commands (/start, /status) to deliver real-time repository notifications to users
- Architected **Firestore** persistence layer mapping Telegram chat IDs to GitHub App installations with per-chat notification preferences (mute/unmute, status, unlink)
- Automated containerized deployment to **Google Cloud Run** via **GitLab CI/CD pipeline**, reducing manual deployment overhead.

TECHNICAL SKILLS

Languages & Frameworks: HTML, Tailwind CSS, React, Python, Express.js, Golang

Technologies: Git/GitHub, GitLab, Linux, Docker, MySQL, MongoDB, Vite, Selenium, Vercel, Firebase, GCP, AWS, Jira, Confluence

LEADERSHIP & INVOLVEMENT

SMU Coder's Assembly | [Article](#)

August 2025 - Present

Coding Mentor

- Mentoring 10+ freshmen in Python and full-stack web development (HTML, JS, Express.js, MongoDB) through weekly sessions.